VIDYA NGO • TECHFEST 2026

Smart Bus Tracking System

A low-cost display solution to bring real-time bus arrival information to rural bus stops

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Add project / bus-stop model image

INTRODUCTION

Finding Problems That Matter

As part of VIDYA NGO's Techfest, we were asked to identify a real community problem and design a practical, low-cost solution for it. We looked around our own surroundings and spoke to daily commuters in rural areas to understand what slows them down every day.

Add team / field-visit photo

Ground Study

Visited local bus stops to observe commuter struggles first-hand.

Community Voice

Spoke with villagers, students & daily workers about their transit issues.

Practical Fix

Focused on a solution buildable with affordable, available hardware.

THE PROBLEM

Rural Areas Have No Bus Tracking Access

Apps like Chalo have made bus tracking easy in cities — but this technology rarely reaches rural areas. At most rural bus stops, commuters have no idea which bus is coming, or when.

The Core Gap

No digital infrastructure exists at rural bus stops to show live bus arrival information — commuters rely entirely on guesswork.

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Real-time tracking apps
usable at most rural stops

Add photo of a rural bus stop

THE PROBLEM

How This Affects Everyday Life

Wasted Waiting Time

Commuters wait for long, uncertain periods without knowing the bus schedule.

Students Miss Classes

Unclear arrival times make it hard for students to plan and reach school on time.

Daily Wage Workers Lose Income

Workers lose paid hours waiting at stops instead of reaching job sites.

No Alternate Planning

Without arrival estimates, people can't decide between walking, waiting, or other transport.

Add image of commuters waiting

THE PROBLEM

Why Apps Like Chalo Don't Reach Rural India

Poor Internet Connectivity

Weak or no mobile data coverage makes real-time apps unreliable.

Limited Smartphone Access

Many rural commuters, especially elderly and daily workers, don't own smartphones.

No GPS-Enabled Buses

Rural bus operators rarely have GPS units fitted, so no live location data exists.

Cost of Data & Devices

Recurring data costs and app dependency are unaffordable for many households.

Result: The digital-tracking revolution simply hasn't reached rural bus stops.

Add comparison image: city app vs rural bus stop

OUR SOLUTION

Introducing the Smart Bus Stop Display

We designed a low-cost Bus Tracking System using a TFT 3.5" display and Arduino UNO, mounted directly at the bus stop. It shows commuters which bus is coming and its expected arrival time — no smartphone or internet needed on the commuter's end.

- Works without the commuter needing internet or a smartphone
- Built using affordable, easily available components
- Mounted directly onto a physical bus stop model / structure
- Displays live bus number and approximate arrival time

Add image/mockup of the display at bus stop

OUR SOLUTION

How the System Works

Arduino UNO

Acts as the processing unit, running the tracking logic.

→

API Call

Bus schedule / arrival data is fed into the Arduino.

→

Processing

Arduino processes and updates arrival info in real time.

→

TFT Display Output

3.5" display shows bus number & expected arrival time.

Add system flow diagram / circuit photo

OUR SOLUTION

Components Used

Add component photo

TFT 3.5" Display

Shows live bus number and estimated arrival time to commuters at the stop.

Add component photo

Arduino UNO

Core microcontroller that runs the logic and drives the display.

Add component photo

Connector Cables

Links the display to the Arduino UNO for data & power transfer.

OUR SOLUTION

Our Working Model

We built a physical model of a rural bus stop with the TFT display mounted onto it, connected to the Arduino UNO, to demonstrate exactly how the final installation would look and function in the field.

- Scaled bus stop structure
- Display fixed at eye level for easy viewing
- Arduino housed safely within the structure

Add photo of the physical bus-stop model / prototype

IMPACT

Benefits for the Community

Saves Time

Commuters can plan their day instead of waiting blindly.

Saves Money

No smartphone or data plan required to use the system.

Helps Students & Workers

Reliable timing helps people reach school and work punctually.

Inclusive for All

Usable by anyone standing at the stop — no tech skill needed.

Add image showing community using the display

WHAT'S NEXT

Future Scope

GPS Integration

Add GPS modules on buses for fully live, accurate tracking instead of scheduled estimates.

IoT Connectivity

Connect multiple stops via IoT to sync real-time data across an entire route.

Companion App

Optional mobile app for commuters who do have smartphone access.

Scalable Rollout

Expand from a single prototype to multiple bus stops across rural regions.

Solar Powered Units

Make each display self-sufficient with solar panels for areas with power gaps.

Govt. & NGO Partnership

Collaborate with transport authorities and NGOs for wider community adoption.

CONCLUSION

A Small Display, A Big Difference

Our Bus Tracking System shows that a simple, affordable idea can solve a daily problem for an entire community — helping people manage their time and travel with confidence, even without internet or smartphones.

Add closing / team photo

Thank You

Shraddha Malve | VIDYA NGO Techfest